UnBake: PBR Material Estimation for 3D Meshes with Baked-In Lighting using Generative Priors

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Abstract

Many scanned and generated 3D assets have lighting baked into their textures, so they cannot be relit correctly. We present UnBake, a training-free method that recovers PBR materials (base color, roughness and metallic) from such assets. UnBake guides the 3D material prior of TRELLIS.2 at test time with two losses: a rebake loss, which requires that the predicted base color, lit by a fitted lighting, reproduces the baked texture, and confidence-weighted material estimates from the 2D model RGBX. The output keeps the input's UV layout. To evaluate the method, we build a dataset of PBR assets paired with versions that have lighting baked into their textures. On this dataset, we compare existing methods in UV space and under relighting with held-out lighting. Unguided TRELLIS.2 recovers the base color most accurately, but neither it nor any other method estimates roughness and metallic with a lower error than a constant guess, which motivates our guidance.

1 Introduction

Scanned and generated 3D assets often have lighting baked into their textures, which prevents them from being correctly relit under novel illumination. Yet robotics simulation, games and content creation, and interior design all require relightable assets with PBR materials (albedo, roughness, metallic). Existing PBR texturing methods largely ignore the input texture and generate new materials from scratch, while 2D intrinsic decomposition models [5] work per view, and often leave baked shadows in the albedo. 3D generative models such as TRELLIS.2 [4] provide a strong prior over materials, but discard the existing texture entirely. In this work, we combine both approaches: we use TRELLIS.2 as a 3D prior that keeps the estimated materials consistent across the whole object, and guide its sampling without any training towards materials that agree with the input asset. Then, fused material estimates from a 2D prior provide targets for the materials, and a rebake loss checks whether these materials, lit by an estimated environment, reproduce the observed baked texture. To test our model, we build a custom dataset that pairs assets with ground-truth PBR materials with baked-lighting meshes generated from selected environment maps. In summary, our contributions are as follows:

- **Dataset.** We curate PBR assets from Objaverse-XL [1] and pair them with baked-lighting counterparts.
- **Method.** We propose a training-free approach that recovers PBR materials by combining 2D intrinsic priors from RGBX with the 3D material prior of TRELLIS.2.

2 Related Work

Recent generative models provide strong priors for material estimation: RGBX [5] predicts intrinsic maps such as albedo, roughness and metallic from single images, while TRELLIS.2 [4] generates

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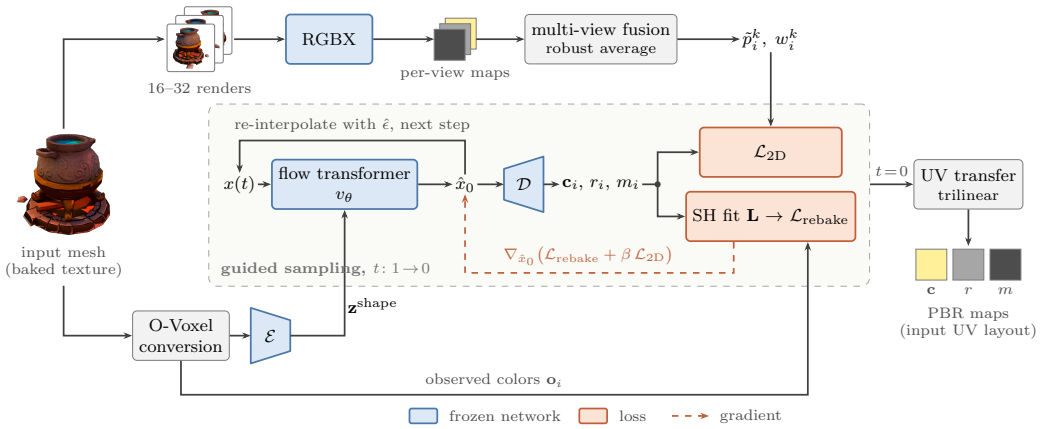


Figure 1: Overview of UnBake. A 2D prior, RGBX, predicts base color, roughness and metallic from renders of the baked input mesh (top). The per-view predictions are averaged into per-voxel estimates $\tilde{p}_i^k$ with confidence weights w_i^k. During sampling, TRELLIS.2 predicts the clean materials $\hat{x}_0$ at selected steps (center). We correct this prediction with the gradient of two losses: $\mathcal{L}_{\text{2D}}$ keeps the decoded materials close to the 2D estimates, and $\mathcal{L}_{\text{rebake}}$ requires that the base color, lit by a fitted lighting $\mathbf{L}$, reproduces the observed baked colors $\mathbf{o}_i$. Finally, the per-voxel materials are transferred into the input's UV layout (right). All networks stay frozen.

PBR materials natively in a sparse 3D latent conditioned on geometry and an image. SuperMat [2] predicts albedo, metallic and roughness in a single diffusion step and fuses multi-view predictions in UV space for 3D objects. Material Anything [3] estimates materials for objects with baked lighting view by view using a trained image diffusion model. We instead combine both priors training-free, guiding TRELLIS.2 [4] with a rebake loss under estimated lighting and fused RGBX [5] estimates, and keep the input UV layout.

3 Data

We curate assets from Objaverse-XL [1], keeping those with non-constant albedo, roughness, and metallic maps. We additionally filter for clean UV maps, keeping only those with less than 1% degenerate faces or overlapping texels. For each asset, Blender Cycles bakes diffuse lighting into the original UV layout under randomly rotated HDR environment maps. The baked lighting includes cast shadows, ambient occlusion and interreflections. The baked texture is the albedo multiplied by the irradiance, exported as an unlit material. Since the geometry is unchanged, this yields pixel-aligned PBR $\Leftrightarrow$ baked-lighting pairs. Our current test dataset has 10 assets, each baked under 8 environment maps, for 80 pairs in total.

4 Method

Given a textured mesh whose texture has lighting and shadows baked in, our goal is to recover base color, roughness and metallic maps in the mesh's original UV layout. Our method is training-free and builds on two frozen priors. First, we fuse multi-view RGBX [5] estimates into per-voxel material priors. Second, we guide the TRELLIS.2 [4] material generator with these priors and a rebake loss under estimated lighting. Finally, we transfer the result back to the input UV layout. As preprocessing, we convert the input mesh to O-Voxel space with TRELLIS.2's conversion routines. We encode the shape features with the frozen shape encoder into the shape latent $\mathbf{z}^{\text{shape}}$, and sample the input texture at each voxel to obtain the observed colors $\mathbf{o}_i$. Figure 1 gives an overview over the proposed method.

2D intrinsic priors. Before sampling, we render 16–32 views of the baked asset under neutral lighting and apply RGBX [5] to each view to predict albedo, roughness and metallic maps, treating the albedo as base color. We project every voxel into each view where it is visible, sample the predicted values, and robustly average them across views to obtain the fused estimates $\tilde{\mathbf{c}}_i$, $\tilde{r}_i$ and

$\tilde{m}_i$, written jointly as $\tilde{p}_i^k$ with $k \in \{\mathbf{c}, r, m\}$. For each channel k, we compute the per-voxel standard deviation σ_i^k across views and turn it into a confidence weight

$$w_i^k = \frac{1}{1 + (\sigma_i^k / \tau^k)^2}$$

where τ^k is the median of σ_i^k over all voxels. Voxels seen in fewer than two views receive zero weight.

Material stage. TRELLIS.2 represents an asset as a sparse voxel grid (O-Voxel), where each active voxel i stores shape features f_i^{shape} and a six-channel PBR feature $f_i^{\text{mat}} = (\mathbf{c}_i, m_i, r_i, \alpha_i)$, i.e. base color, metallic, roughness and opacity. Material latents are generated by a sparse rectified flow transformer, conditioned on the shape latents and on DINOv3 features of the input image via cross-attention. The forward process interpolates linearly, $x(t) = (1-t)\, x_0 + t\, \epsilon$, and the network $v_\theta(x(t), t)$ is trained to predict $\epsilon - x_0$. Sampling integrates v_θ from $t = 1$ to $t = 0$.

Guided sampling. At selected steps t of the rectified flow sampler, we estimate the clean latent $\hat{x}_0 = x(t) - t\, v_\theta(x(t), t)$, which follows from $x(t) = x_0 + t\,(\epsilon - x_0)$. We decode $\hat{x}_0$ with the frozen material decoder $\mathcal{D}$ into per-voxel base color $\mathbf{c}_i$, roughness r_i and metallic m_i. Let $\mathbf{o}_i$ denote the observed (baked) color of voxel i, sampled from the input texture. We model it as Lambertian shading under second-order Spherical Harmonics (SH) lighting $\mathbf{L} \in \mathbb{R}^{9\times 3}$:

$$\hat{\mathbf{o}}_i = \mathbf{c}_i \odot (\mathbf{t}_i^\top \mathbf{L})$$

.

Here $\mathbf{t}_i \in \mathbb{R}^9$ is the SH projection of the cosine-weighted visibility, precomputed from the geometry to account for baked shadows. Since $\hat{\mathbf{o}}_i$ is linear in $\mathbf{L}$, we fit $\mathbf{L}$ by least squares at each guidance step. The rebaked colors are compared to the observations, and the decoded materials p_i^k to the fused 2D estimates $\tilde{p}_i^k$:

$$\mathcal{L}_{\text{rebake}} = \frac{1}{N} \sum_{i=1}^{N} \text{Huber}\left(\hat{\mathbf{o}}_i - \mathbf{o}_i\right), \qquad \mathcal{L}_{\text{2D}} = \sum_{k \in \{\mathbf{c}, r, m\}} \lambda_k \frac{\sum_i w_i^k \left\| p_i^k - \tilde{p}_i^k \right\|_1}{\sum_i w_i^k}.$$

Together wit the prior, the rebake loss separates base color from baked lighting (on its own it cannot fix the overall brightness of the base color, since lighting is refit), while roughness and metallic are constrained only by $\mathcal{L}_{\text{2D}}$. We update $\hat{x}_0 \leftarrow \hat{x}_0 - \eta\, \nabla_{\hat{x}_0}(\mathcal{L}_{\text{rebake}} + \beta\, \mathcal{L}_{\text{2D}})$ by backpropagating through $\mathcal{D}$ only. We then continue sampling from the corrected estimate by re-interpolating with the predicted noise $\hat{\epsilon} = x(t) + (1-t)\, v_\theta(x(t), t)$, i.e. $x(t) \leftarrow (1-t)\, \hat{x}_0 + t\, \hat{\epsilon}$. All network weights remain frozen.

Output. Finally, we transfer the decoded per-voxel materials back to the input mesh's original UV layout. For each texel of the input texture, we compute the 3D surface point it covers and sample $\mathbf{c}_i$, r_i and m_i there by trilinear interpolation of the neighboring voxel attributes. The resulting base color, roughness and metallic maps have the same resolution and UV layout as the input texture, so they can replace it directly without re-unwrapping or modifying the mesh.

5 Experiments

In this section, we report baseline metrics for input as albedo, SuperMat [2], Material Anything [3], and unguided TRELLIS.2 [4] at 512^3 and 1024^3 resolution. Input as albedo simply takes the observed color as albedo and sets roughness to a constant value 0.5 and metallic to 0.0. We report per-channel MAE and PSNR against ground-truth materials, and MAE, PSNR, SSIM, and LPIPS on renders relit under novel lighting using 3 held-out HDRI environment maps. All metrics are averaged over all 80 items of our custom baked-lighting dataset. For relighting, we sample 8 different views and average metrics across views. MAE in UV space is additionally weighted by surface area. For time reasons, only Material Anything was tested on a 20 sample subset of the dataset.

Table 1: Material accuracy in UV space, averaged over 80 dataset items (20 item subset for Material Anything). Best values in bold.

	Albedo		Roughness		Metallic	
Method	MAE ↓	PSNR ↑	MAE ↓	PSNR ↑	MAE ↓	PSNR ↑
Input as albedo	0.151	13.30	**0.204**	**12.05**	**0.238**	7.38
TRELLIS.2 (unguided, 512^3)	0.123	15.55	0.349	7.52	0.349	5.78
TRELLIS.2 (unguided, 1024^3)	**0.115**	**15.87**	0.418	6.40	0.435	4.50
SuperMat	0.157	13.87	0.219	11.13	0.268	**8.73**
Material Anything	0.262	9.98	0.238	10.71	0.407	5.48

Table 2: Relighting under 3 held-out environment maps, 8 views each, averaged over 80 dataset items (20 item subset for Material Anything). Best values in bold.

Method	LPIPS ↓	PSNR ↑	SSIM ↑	MAE ↓
Input as albedo	0.070	14.42	**0.750**	0.165
TRELLIS.2 (unguided, 512^3)	0.087	14.49	0.620	0.155
TRELLIS.2 (unguided, 1024^3)	0.085	14.09	0.585	0.172
SuperMat	**0.068**	**15.62**	0.740	**0.149**
Material Anything	0.094	12.49	0.607	0.204

Results. TRELLIS.2 recovers the base color best (Table 1): at 1024^3 its MAE is 24% lower than input as albedo, and it removes the color tints that the other methods copy from the input (Fig. 2). However, no method beats the constant baseline on roughness and metallic. Most surfaces in our dataset are not metal, yet TRELLIS.2 predicts metallic and roughness values close to 1 almost everywhere for the cauldron (Fig. 3). SuperMat gives the best relighting results (Table 2). TRELLIS.2 is clearly behind in relighting because it wrongly predicts metal and roughness: its surfaces reflect the environment like mirrors, which lowers all relighting metrics. Material Anything is worst in both tables because its base color is far too bright, and this also holds on its 20-sample subset. In summary, the 3D prior gives the best base color, but roughness and metallic need additional guidance.

6 Discussion

To our knowledge, this is the first method that couples a 3D-native generative material prior with a physically based inverse-rendering loss at test time. The lighting is fitted jointly in closed form, so neither the illumination nor training data is needed, and the output keeps the input's UV layout.

Our baselines support this design. TRELLIS.2 already gives the most accurate base color, and it is consistent across the whole object. The rebake loss additionally ties it to the input texture and accounts for baked shadows by modeling which parts of the surface are occluded. Moreover, the material stage takes only minutes per asset.

The main limitation concerns roughness and metallic. Our dataset bakes only diffuse lighting, which does not depend on these two channels, so the rebake loss cannot correct them. They therefore rely entirely on the priors, which so far do not beat a constant guess. If RGBX also predicts too much metal, as TRELLIS.2 does, we may need to bias metallic towards 0 or model specular reflections in the rebake loss. Second, our simple lighting model cannot reproduce sharp lighting details or light bouncing between surfaces, so these remain in the base color. Third, strong guidance can push the sample towards materials the prior would not produce, and details smaller than a voxel are lost. Finally, 10 assets is a small dataset. Next, we will implement the guided sampling, test the effect of each component, and recover fine texture details by dividing the input texture by the estimated lighting.

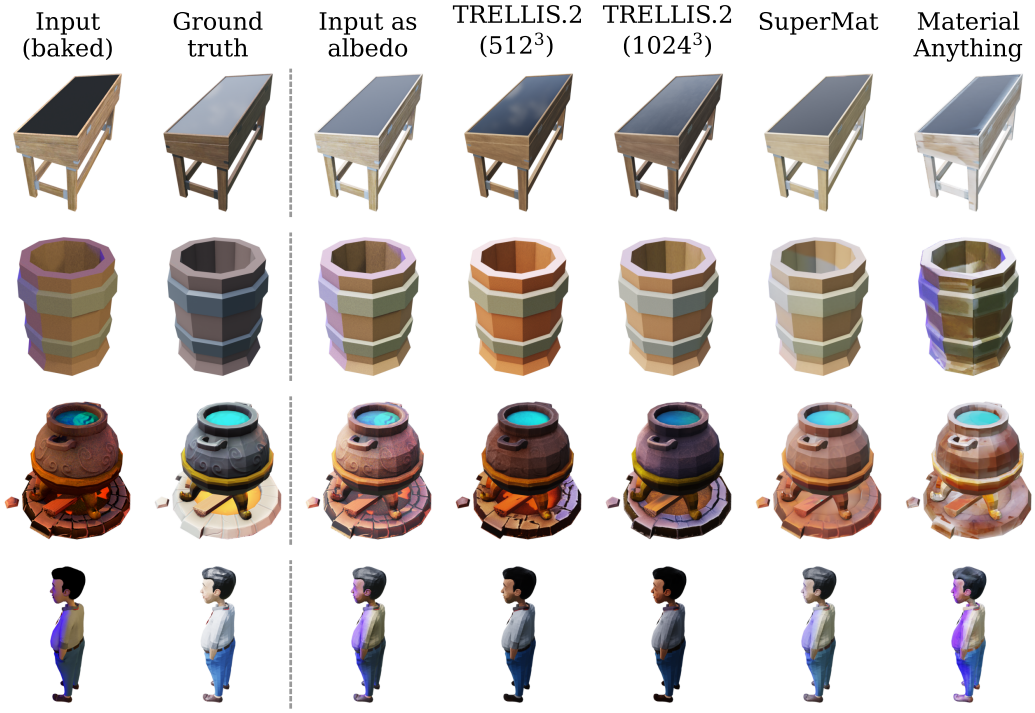


Figure 2: Relighting results. Each row shows one object: its baked input, the ground truth, and each method's predicted materials.

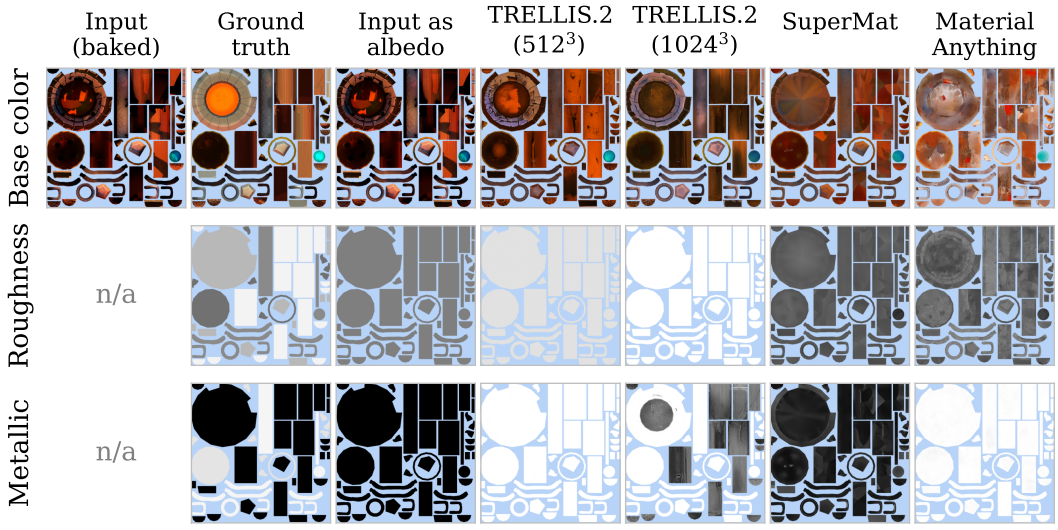


Figure 3: Predicted material maps for the cauldron in Fig. 2, shown in the ground truth's UV layout. Roughness and metallic are in grayscale (black $= 0.0$, white $= 1.0$) and texels outside the UV islands are in blue. The input has no roughness or metallic maps.

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